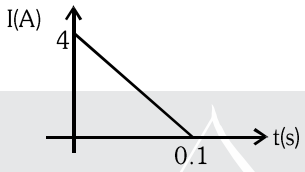
SYLLABUS : ELECTROMAGNETIC INDUCTION

1. When magnetic flux through a coil is changed, the variation of induced current in the coil with time is as shown in graph. If resistance of coil is 10 Ω, then the total change in flux of coil will be–

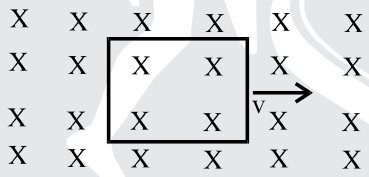


(A) 4 (B) 8 (C) 2 (D) 6

2. A coil having 500 square loops each of side 10 cm is placed normal to a magnetic field which increased at a rate of 1.0 T/sec. The induced e.m.f. (in volt) is :–

(A) 0.1 (B) 0.5 (C) 1.0 (D) 5.0

3. A square frame of wire of side l and resistance R moves in its plane with a uniform velocity v perpendicular to one of its sides. A uniform and constant magnetic field B exists along the perpendicular to the plane of the loop in fig. The current induced in the 3loop is -



(A) $2Bl\upsilon/R$ anticlockwise (B) $Bl\upsilon/R$ anticlockwise

(C) $Bl\upsilon/R$ clockwise (D) zero

4. A small, circular loop of wire is placed inside a long solenoid carrying a current. The plane of the loop contains the axis of the solenoid. If the current in the solenoid is varied, the current induced in the loop is

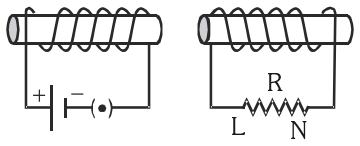
(A) anticlockwise

(B) clockwise

(C) zero

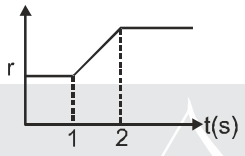
(D) clockwise or anticlockwise depending on whether the resistance in increased or decreased.

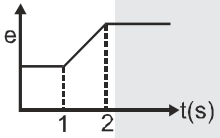
5. Two co–axial solenoids shown in figure. If key of primary suddenly opened then direction of instantaneous induced current in resistance 'R' which connected in secondary :–



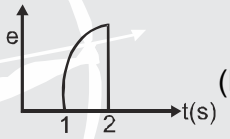
(A) L to N (B) N to L (C) alternating (D) zero

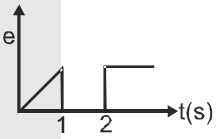
6. A flexible wire bent in the form of a circle is placed in a uniform magnetic field perpendicular to the plane of the coil. The radius of the coil changes with time as shown figure. The induced emf in the coil is :



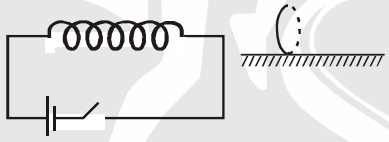
(A) 

(B) 

(C) 

(D) 

7. A horizontal solenoid is connected to a battery and a switch (figure). A conducting ring is placed on a frictionless surface, the axis of the ring being along the axis of the solenoid. As the switch is closed, the ring will



(A) move towards the solenoid

(B) remain stationary

(C) move away from the solenoid

(D) move towards the solenoid or away from it depending on which terminal (positive or negative) of the battery is connected to the left end of the solenoid.

8. Two identical coaxial circular loops carry a current i each circulating in the same direction. If the loops approach each other

(A) the current in each loop will decrease

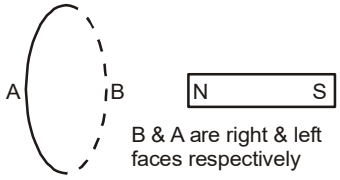
(B) the current in each loop will increase

(C) the current in each loop will remain the same

(D) the current in one loop will increase and in the other loop will decrease

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9. In the figure shown, the magnet is pushed towards the fixed ring along the axis of the ring and it passes through the ring.



(A) when magnet goes towards the ring the face B becomes south pole and the face A becomes north pole

(B) when magnet goes away from the ring the face B becomes north pole and the face A becomes south pole

(C) when magnet goes away from the ring the face A becomes north pole and the face B becomes south pole

(D) the face A will always be a north pole.

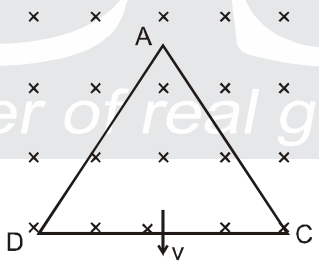
10. A wire of length ℓ is moved with a constant velocity $\vec{\upsilon}$ in a magnetic field. A potential difference appears across the two ends

(A) if $\vec{\upsilon} \,||\, \vec{\ell}$ (B) if $\vec{\upsilon} \,||\, \vec{B}$ (C) if $\vec{\ell} \,||\, \vec{B}$ (D) none of these

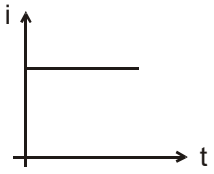
11. A uniform magnetic field exists in region given by $\vec{B} = 3\hat{i} + 4\hat{j} + 5\hat{k}$. A rod of length 5 m is placed along $y-$axis is moved along $x-$axis with constant speed 1 m/sec. Then induced e.m.f. in the rod will be:

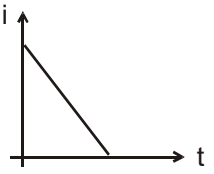
(A) zero (B) 25 v (C) 20 v (D) 15 v

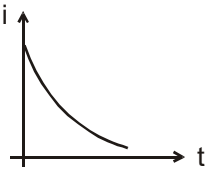
12. An equilateral triangular loop ADC having some resistance is pulled with a constant velocity v out of a uniform magnetic field directed into the paper. At time t = 0, side DC of the loop is at edge of the magnetic field. The induced current (i) versus time (t) graph will be as

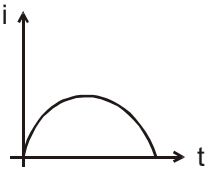


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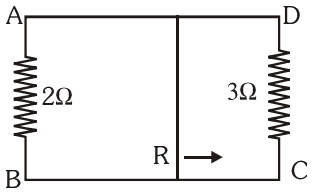
(A) 

(B) 

(C) 

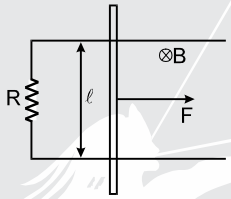
(D) 

13. A rectangular loop with a sliding connector of length 10 cm is situated in uniform magnetic field perpendicular to plane of loop. The magnetic induction is 0.1 tesla and resistance of connector (R) is 1 ohm. The sides AB and CD have resistances 2 ohm and 3 ohm respectively. Find the current in the connector during its motion with constant velocity one metre / sec.



(A) $\frac{1}{110}$A (B) $\frac{1}{220}$A (C) $\frac{1}{55}$A (D) $\frac{1}{440}$A

14. A constant force F is being applied on a rod of length ' ℓ ' kept at rest on two parallel conducting rails connected at ends by resistance R in uniform magnetic field B as shown.



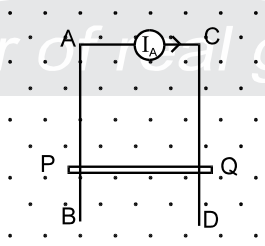
(A) the power delivered by force will be constant with time

(B) the power delivered by force will be increasing first and then it will decrease

(C) the rate of power delivered by the external force will be increasing continuously

(D) the rate of power delivered by external force will be decreasing continuously before becoming zero.

15. AB and CD are fixed conducting smooth rails placed in a vertical plane and joined by a constant current source at its upper end. PQ is a conducting rod which is free to slide on the rails. A horizontal uniform magnetic field exists in space as shown. If the rod PQ is released from rest then,



(A) The rod PQ may move downward with constant acceleration

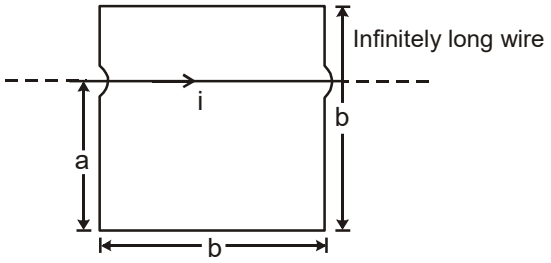
(B) The rod PQ may move upward with constant acceleration

(C) The rod will move downward with decreasing acceleration and finally acquire a constant velocity

(D) either A or B.

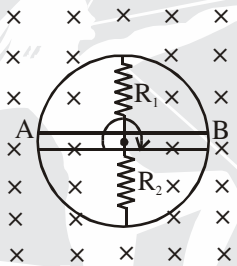
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16. For the situation shown in the figure, flux through the square loop is :



(A) $\left(\frac{\mu_0 ia}{2\pi}\right)\ell n\left(\frac{a}{2a-b}\right)$ (B) $\left(\frac{\mu_0 ib}{2\pi}\right)\ell n\left(\frac{a}{2b-a}\right)$ (C) $\left(\frac{\mu_0 ib}{2\pi}\right)\ell n\left(\frac{a}{b-a}\right)$ (D) $\left(\frac{\mu_0 ia}{2\pi}\right)\ell n\left(\frac{2a}{b-a}\right)$

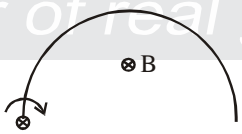
17. AB is resistanceless conducting rod which forms a diameter of a conducting ring of radius r rotating in a uniform magnetic field B as shown. The resistors R_1 and R_2 do not rotate. Then current through the resistor R_1 is–



(A) $\frac{B\omega r^2}{2R_1}$ (B) $\frac{B\omega r^2}{2R_2}$ (C) $\frac{B\omega r^2}{2R_1R_2}(R_1+R_2)$ (D) $\frac{B\omega r^2}{2(R_1+R_2)}$

18. A semicircular wire of radius R is rotated with constant angular velocity ω about an axis passing through one end and perpendicular to the plane of the wire. There is a uniform magnetic field of strength B. The induced e.m.f. between the ends is–

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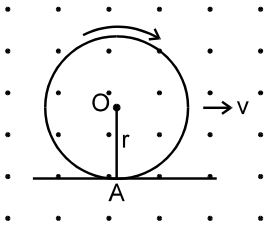


(A) B $\omega R^2/2$ (B) 2B ωR^2 (C) is variable (D) none of these

19. A conducting rod of length ℓ rotates with a uniform angular velocity ω about its perpendicular bisector. A uniform magnetic field B exists parallel to the axis of rotation. The potential difference between the two ends of the rod is

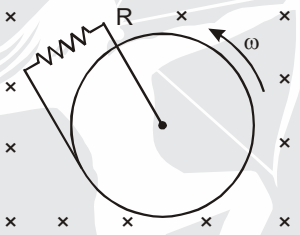
(A) $2B\omega\ell^2$ (B) $\frac{1}{2}\omega B\ell^2$ (C) $B\omega\ell^2$ (D) zero

20. A conducting ring of radius r with a conducting spoke is in pure rolling on a horizontal surface in a region having a uniform magnetic field B as shown, v being the velocity of the centre of the ring. Then the potential difference $V_0 - V_A$ is



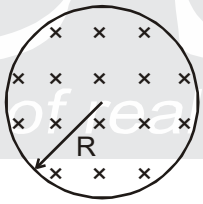
(A) $\frac{Bvr}{2}$ (B) $\frac{3Bvr}{2}$ (C) $\frac{-Bvr}{2}$ (D) $\frac{3Bvr}{2}$

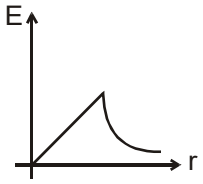
21. Figure shows a conducting disc rotating about its axis in a perpendicular uniform and constnat magnetic field B. A resistor of resistance R is connected between the centre and the rim. The radius of the disc is 5.0 cm, angular speed ω = 40 rad/s, B = 0.10 T and R = 1 Ω. The current through the resistor is

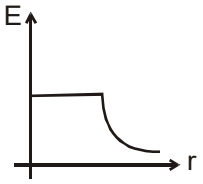


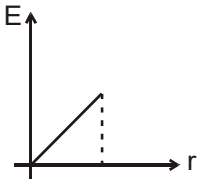
(A) 5 mA (B) 50 A (C) 5 A (D) 10 mA

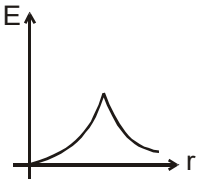
22. A cylindrical space of radius R is filled with a uniform magnetic induction B parallel to the axis of the cylinder. If B changes at a constant rate, the graph showing the variation of induced electric field with distance r from the axis of cylinder is



(A) 

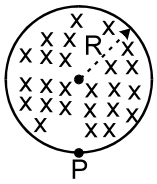
(B) 

(C) 

(D) 

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23. A uniform magnetic field of induction B is confined to a cylindrical region of radius R. The magnetic field is increasing at a constant rate of $\frac{dB}{dt}$ (Tesla/second). An electron of charge q, placed at the point P on the periphery of the field experiences an acceleration :



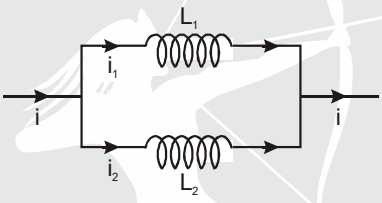
(A) $\frac{1}{2}\frac{eR}{m}\frac{dB}{dt}$ toward left

(B) $\frac{1}{2}\frac{eR}{m}\frac{dB}{dt}$ toward right

(C) $\frac{eR}{m}\frac{dB}{dt}$ toward left

(D) zero

24. Two inductors L_1 and L_2 are connected in parallel and a time varying current i flows as shown. The ratio of currents i_1/i_2 at any time t is



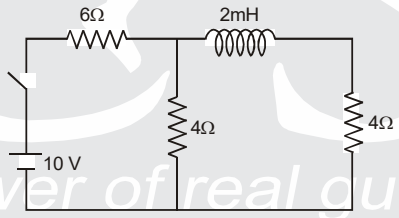
(A) L_1/L_2

(B) L_2/L_1

(C) $\frac{L_1^2}{(L_1+L_2)^2}$

(D) $\frac{L_2^2}{(L_1+L_2)^2}$

25. In the given circuit find the ratio of i_1 to i_2. Where i_1 is the initial (at t = 0) current, and i_2 is steady state (at t = ∞) current through the battery :



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(A) 1.0

(B) 0.8

(C) 1.2

(D) 1.5

ANSWER KEY

1.	(C)	**2.**	(D)	**3.**	(D)	**4.**	(C)	**5.**	(A)
6.	(B)	**7.**	(D)	**8.**	(A)	**9.**	(C)	**10.**	(D)
11.	(B)	**12.**	(B)	**13.**	(B)	**14.**	(D)	**15.**	(D)
16.	(C)	**17.**	(A)	**18.**	(B)	**19.**	(D)	**20.**	(C)
21.	(A)	**22.**	(A)	**23.**	(A)	**24.**	(B)	**25.**	(B)